

GROUP BY / ORDER BY Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE.

AIM: To group query results using GROUP BY and sort them using ORDER BY.

-- 1. ORDER BY: Sort results

```
-- Ascending order (default) by age
SELECT FacName, Age FROM FACULTY ORDER BY Age;

-- Descending order
SELECT FacName, Age FROM FACULTY ORDER BY Age DESC;
```

```
mysql> SELECT FacName, Age FROM FACULTY ORDER BY Age;
+-----+-----+
| FacName | Age |
+-----+-----+
| Sunitha | 46 |
| Ramesh  | 48 |
| Arjun   | 51 |
| Mohan   | 59 |
| Kavya   | 68 |
+-----+-----+
5 rows in set (0.02 sec)
```

```
-- Sort by multiple columns
SELECT * FROM STUDENT ORDER BY City ASC, Name ASC;
```

```
mysql> -- Sort by multiple columns
mysql> SELECT * FROM STUDENT ORDER BY City ASC, Name ASC;
+-----+-----+-----+-----+-----+-----+
| RegNo | Name          | Gender | DOB          | MobileNo | City          |
+-----+-----+-----+-----+-----+-----+
| 191   | Aravind Kumar | M      | 2001-04-12   | 9884012345 | Chennai      |
| 193   | Karthik       | M      | 2000-12-05   | 9884012347 | Coimbatore   |
| 192   | Divya         | F      | 2001-07-19   | 9884012346 | Madurai      |
| 194   | Meena         | F      | 2001-02-27   | 9884012348 | Salem       |
| 195   | Suresh        | M      | 2000-08-16   | 9884012349 | Trichy       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

-- 2. GROUP BY: Aggregate rows into groups

```
-- Count faculty per department
SELECT Dept, COUNT(*) AS num_faculty
FROM FACULTY
GROUP BY Dept;
```

```
mysql> -- Count faculty per department
mysql> SELECT Dept, COUNT(*) AS num_faculty
      -> FROM FACULTY
      -> GROUP BY Dept;
+-----+-----+
| Dept | num_faculty |
+-----+-----+
| CSE  |          3 |
| ECE  |          1 |
| IT   |          1 |
+-----+-----+
3 rows in set (0.04 sec)

mysql> |
```

```
-- Average age of faculty per department
SELECT Dept, AVG(Age) AS avg_age
FROM FACULTY
GROUP BY Dept;
```

```
mysql> SELECT Dept, AVG(Age) AS avg_age
      -> FROM FACULTY
      -> GROUP BY Dept;
+-----+-----+
| Dept | avg_age |
+-----+-----+
| CSE  | 52.6667 |
| ECE  | 46.0000 |
| IT   | 68.0000 |
+-----+-----+
3 rows in set (0.02 sec)
```

-- 3. GROUP BY with HAVING: Filter groups

```
-- Departments with more than 1 faculty member
SELECT Dept, COUNT(*) AS num_faculty
FROM FACULTY
GROUP BY Dept
HAVING COUNT(*) > 1;
```

```
mysql> SELECT Dept, COUNT(*) AS num_faculty
-> FROM FACULTY
-> GROUP BY Dept
-> HAVING COUNT(*) > 1;
+-----+-----+
| Dept | num_faculty |
+-----+-----+
| CSE  |          3 |
+-----+-----+
1 row in set (0.02 sec)
```

-- 4. GROUP BY + ORDER BY combined

```
-- Group faculty count per department, sorted highest first
SELECT Dept, COUNT(*) AS num_faculty
FROM FACULTY
GROUP BY Dept
ORDER BY num_faculty DESC;
```

```
mysql> SELECT Dept, COUNT(*) AS num_faculty
-> FROM FACULTY
-> GROUP BY Dept
-> ORDER BY num_faculty DESC;
+-----+-----+
| Dept | num_faculty |
+-----+-----+
| CSE  |          3 |
| ECE  |          1 |
| IT   |          1 |
+-----+-----+
3 rows in set (0.00 sec)
```